

Tab 1

Team directory

 Picture	 Name	 Role	 Location
NO	Tanay	Engineering	Place
NO	Arnav	Engineering	Place
NO	Ishan	Project management	Place
NO	Chris	Engineering	Place
NO	Nichelle	Engineering	Place

RACI Chart

https://lucid.app/lucidspark/8407b481-a98e-4d09-90b6-255d3a008a6b/edit?page=0_0&invitationId=inv_a8c14cd8-0602-488f-a823-f15a3db270c6#

Template Entry

Date:[DATE][SIGNATURE]

Who is Present: [INSERT NAME HERE]

What did you accomplish at the meeting?:

Important decisions:

Record any group decisions made about the project

Action Items/Goals:

With due dates

Notes, etc:

Background

Matrices

Comments:

Other important info

Date: 2/23/2026

Who is Present: Ishan, Chris, Tanay, Nichelle, Arnav

What did you accomplish at the meeting?: We discussed issues faced by people of advanced age and discussed possible ideas

Important decisions:

An important decision that we have made is that our goal is to focus on people of age or as someone had carefully worded “*We want to focus on people who have more wisdom, to say, or the ones who have seen the world for the longest*”.

Additionally, we want this project to be engineering focused, simple for users, and be able to be fastly iterable.

Action Items/Goals:

The action items and goals before the next meeting are to try and think of an idea before the next meeting, with 3 being an ideal amount. So, the per person list would include:

- Tanay: Think of 3 unique ideas
- Ishan: Think of 3 unique ideas
- Nichelle: Think of 3 unique ideas
- Arnav: Think of 3 unique ideas
- Chris: Think of 3 unique ideas

Notes, etc:

We have made this list of problems that older people, on average, face.

- Difficulty navigating uneven or poorly designed environments
- Vision and Hearing challenges affecting daily tasks
- Managing multiple chronic diseases
- Misunderstanding health instructions
- Limited adoption of health and communication technologies
- Healthcare access bottlenecks
- Lack of provider geriatric training
- Social isolation and loneliness
- Undiagnosed depression and anxiety
- Digital usability problems
- Digital and health literacy gaps
- Influence of social determinants on aging outcomes

- High healthcare and long-term care costs
- Financial insecurity and debt pressures
- Caregiver coordination challenges
- Transportation and mobility barriers
- Inequities in neighborhood design
- Vulnerability to emergency injuries

The three main that our group would be leaning towards would be Difficulty navigating uneven or poorly designed environments, Vision and Hearing challenges affecting daily tasks, and Managing multiple chronic diseases.

Comments:

Other important info

Template Entry**Date: 2/24/2026****Who is Present: [Tanay, Nichelle, Ishan]****What did you accomplish at the meeting?:**

We have decided that we will be working with older people and we wanted to help people with either some motion issues or sleep issues. We also decided on the core components that we would be comfortable with and some that we want to explore.

Important decisions:

Still doing the wise among us, the ones who have lived on earth for longer.

Action Items/Goals:

Now that we have gotten the chance to see others ideas and become exposed to what other group members might be interested in, try and create one final idea, which will be shared at the next meeting:

- Tanay: Think of 1 finalized idea
- Ishan: Think of 1 finalized idea
- Nichelle: Think of 1 finalized idea
- Arnav: Think of 1 finalized ideas
- Chris: Think of 1 finalized idea

Notes, etc:

These were the ideas that we had for the goal from the previous meeting

Tanay's ideas:

- Potential Idea: The system uses camera and IMU data to predict balance instability in older adults and applies low-level galvanic vestibular stimulation to bias postural perception and reduce fall risk, rather than attempting real-time physical correction
- Buttons in real life that correspond to buttons on the phone
- Galvanic Vestibular Stimulation for sleep

Ishan's ideas:

- Old people physical therapy trainer
- Phone case that makes text appear more visible for old people
- Shape adapting attachment for handles for old people with arthritis
- Ankle correction module with Reinforcement Learning

Nichelle's ideas:

- Inexpensive but more mobile soft-exterior therapy robot
- ADHD-responsive study environment

- Anxiety-responsive wristband or glove for emotional comfort
- Center-of-gravity-adjusting backpack for balance assistance
- Modular exoskeleton for motion assistance
- Physiologically adaptive soft wearable

Chris's ideas:

- Directional hearing aids
- Cup/mug that doesn't spill when tremors

Comments:

NA

Date: 2/25/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

During this meeting we discussed our central ideas and at this moment the key preliminary idea is using Galvanic Vestibular Stimulation, primarily for correcting people's posture as they sleep, so we started to look into the feasibility of the idea.

Important decisions:

At this moment, put a hold on brainstorming, just be aware of ideas as they hit you as now the main idea is to try and see if GVS for sleep is viable.

Action Items/Goals:

The main goal before the next meeting is to understand how GVS works and see if GVS is feasible for posture correction during sleep.

- Tanay: Learn more about GVS and if it works for sleep.
- Ishan: Learn more about GVS and if it works for sleep.
- Nichelle: Learn more about GVS and if it works for sleep.
- Arnav: Learn more about GVS and if it works for sleep.
- Chris: Learn more about GVS and if it works for sleep.

Notes, etc:

These were the best ideas according to our team after discussing them.

First Idea: Using Galvanic Vestibular Stimulation to correct posture when people sleep.

Second Idea: Device that amplifies noise (hearing aids) based on the direction that the old person is facing

Third Idea: A phone case which corrects for tremors, where it moves to adjust for how people type.

Comments:

NA

Date: 2/27/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

After doing some preliminary research, we have realized that the idea of using GVS for posture correction during sleep is not very useful and suitable. Now, we have started to look at the idea of using GVS for preventing falls in older people.

Task List:

- Chris, Arnav: BoM(b) based solely on the GVS part of it without the electronics to detect anything yet.
- Tanay, Ishan: Seeing what other ideas could be reconsidered.
- Nichelle: Reaching out to clients.

Important decisions:

We have decided that our ideas will involve using Galvanic Vestibular Stimulation but not for sleep. We are currently exploring other uses for Galvanic Vestibular Stimulation.

Notes, etc:

NA

Comments:

NA

Date: 3/2/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

Task List: Finalize the idea and start to narrow the scope for clients and the Bill of Materials

Important decisions:

Finalized the idea of using GVS for preventing falls in those who have more wisdom than we do.

Action Items/Goals:

Find client by 3/20/2026

Finish BOM by 3/18/2026

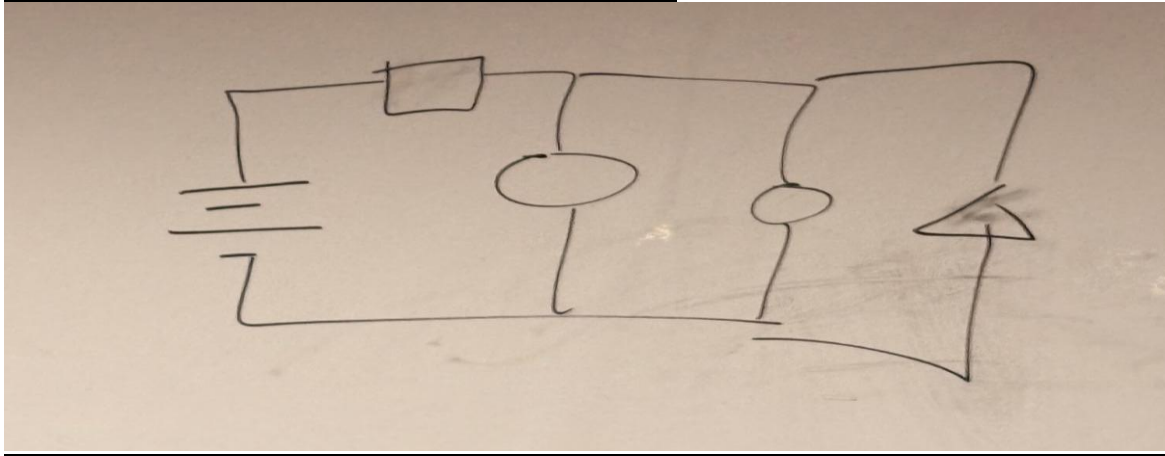
Comments:

The project will be very heavy on the electrical engineering side so people will need to learn the necessary skills.

Date: 3/16/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:



Important decisions:

NOT REALLYt

Action Items/Goals:

Chris, Ishan: BOM

Nichelle: Client

Tanay + Arnav: PDR

Notes, etc:

NA

Action Items/Goals:

BOM due by 3/18

Notes, etc:

Have leads at assistive living centers to find a client

Nothing else :)

FIRE

Date: 3/17/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

BOM HAS BEEN DONE, v1

Also letter has been sent out.

Assistive Technology Project



Nichelle Thinagar <nichelle.thinagar@gmail.com>

to jmadden, chris.chenghao.xu, arnavp2027, ishan.kasam06, mysticalrubi

9:38 AM (0 minutes ago)



Hello Miss Madden,

My name is Nichelle Thinagar, and I am a junior at the Massachusetts Academy of Math and Science. My classmates and I are currently working on a project to prevent falls for individuals with difficulty balancing, and I was directed to you by my friend Khloe, who recently started working at Southgate. Our envisioned device uses something called galvanic vestibular stimulation (GVS), which uses surface electrodes to stimulate the nerve responsible for balance in order to correct for imminent falls. Although we will be testing the actual device on ourselves, we wanted to speak with someone who could benefit from such a device to develop it based on their needs and preferences. As such, we were wondering if any of your residents would be willing to meet with us (over Zoom or in person) to discuss our project.

We greatly appreciate your help!

--
Thanks,
Nichelle

Important decisions:

Decided on the electrodes and resistors to use.

Found test subject.

Action Items/Goals:

Chris, Ishan: BOM

Tanay + Arnav + Nichelle: PDR

Notes, etc:

NA

Action Items/Goals:

WORK ON FINDING MATERIALS FROM OWN HOMES

Notes, etc:

Have leads at assistive living centers to find a client

Date: 3/18/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We started working on the grant, the background and market research section have been mapped out. We have found some sources and are trying to find issues with our pdf.

Important decisions:

We have decided on the fact that we are going to be going with an ESP and going to be using electrodes and IMU's to detect stuff.

Action Items/Goals:

Chris, Ishan: Grant for the BOM

Tanay + Nichelle: PDR

Arnav: Fixing codes and schematic to include IMU module

Notes, etc:

NA

Action Items/Goals:

WORK ON FINDING MATERIALS FROM OWN HOMES

Grant

PDR

Notes, etc:

Have leads at assistive living centers to find a client

Date: 3/19/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

The grant we have done the schematic for the first prototype, we finished writing all the sections of the grant. For the PDR, we have done massive strides on the separate sections and we want finalize the Statement of Problem, Background and Market Research, Target Audience, Team Qualifications and Strengths.

Important decisions:

We have decided on the fact that we are going to be going with an ESP and going to be using electrodes and IMU's to detect stuff.

Action Items/Goals:

Chris, Ishan: Grant for the BOM

Tanay + Nichelle: PDR

Arnav: Fixing codes and schematic to include IMU module

Notes, etc:

NA

Action Items/Goals:

WORK ON FINDING MATERIALS FROM OWN HOMES

Nichelle: Finalizing the Market Research

Tanay: Background Research Finalizing

Arnav: Team Qualifications and Strengths and reviewing Target Audience.

Notes, etc:

Have leads at assistive living centers to find a client

Date: 3/23/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

The grant and the BOM are done. For the PDR, we have done massive strides on the separate sections and we want to finalize the Statement of Problem, Background and Market Research, Target Audience, Team Qualifications and Strengths.

Important decisions:

We have decided on the fact that we are going to be going with an ESP and going to be using electrodes and IMU's to detect stuff.

Action Items/Goals:

FIRST DRAFT DUE ON 3/27/2026

Nichelle: Complete the three designs for the PDR: Make sketches

Tanay: Do the background research section

Ishan: Design requirements, timeline for PDR

Chris: Current solutions for the PDR

Arnav: Team strengths and resources, and the needed budget section for PDR

Notes, etc:

Tanay needs to reach out to his contact at an assistive living center, and if that does not work out, then we will have Nichelle's grandfather as the client.

Date: 3/24/2026

Who is Present: Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We worked on the pdr, taking sections and individually working on them.

Important decisions:

We have to “lock-in” and finish the pdr as soon as possible.

Action Items/Goals:

FIRST DRAFT DUE ON 3/27/2026

Some people are going to be working on writing up the pdr, others will work on the presentation.

Notes, etc:

Date: 3/25/2026

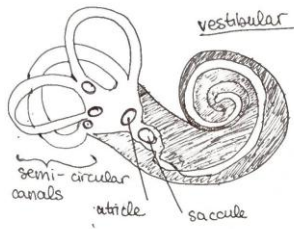
Who is Present: Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We worked on the pdr, taking sections and individually working on them.

#1 Galvanic Vestibular Stimulation (with IMUs)

vestibular system



- semi-circular canals detect rotational motion
- utricle and saccule detect linear acceleration and head tilt
- Signals from sensory organs fail to arrive or arrive corrupted in elderly individuals.

instability detected by both vestibular system and IMU



GVS stimulates the vestibular system by passing a current between two electrodes (anode and cathode) to stimulate.

"criss-cross" setup of electrode: allows for nuanced yaw control

Device may take non-intrusive forms such as sunglasses (above) or headbands to maximize user comfort

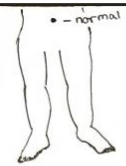
Uses 4-pole GVS

Surface electrodes for minimal intrusiveness

The magnitude of induced correction has a direct relationship with applied amps

#2

- normal center of gravity



- Lowering center of gravity aids balance

- Handheld devices such as this weighted beam are cumbersome (high energy cost)

- Ships use ballasts to maintain balance. Those ballasts fill with air or water to change the buoyant force acting on the vessel.



- A similar system can be applied to human balance where "ballast tanks" with a fluid are suspended on either side of an individual in the coronal plane

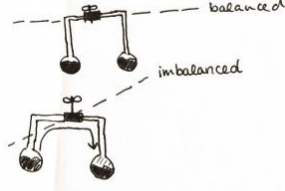


- Uses IMUs to predict coronal plane instability and determine appropriate counterbalancing

- Uses high-speed pumps to move fluid between tanks for balance correction

balanced

imbalanced



Important decisions:

We have to "lock-in" and finish the pdr as soon as possible. We have

Action Items/Goals:

FIRST DRAFT DUE ON 3/27/2026

Some people are going to be working on writing up the pdr, others will work on the presentation.

Notes, etc:

Date: 3/26/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We worked on making the whole thing, finished with a PDR which is good enough to submit but have to go through and submit it. Our presentation is coming along, still need to put in some content.

Important decisions:

There were no important decisions made today.

Action Items/Goals:

Read your respective section of the PDR and finish the presentation.

Notes, etc:

Date: 3/30/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We finished the PDR and the presentation and we went over to it together to make sure it was good.

Important decisions:

Nothing important was decided on today.

Action Items/Goals:

Some people are now going to work on the prototype and the others will reinforce the pdr with Ishan Kasam submitting it.

Notes, etc:

Date: 4/1/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

Some of the prototype is done, testing out the thing with the transistors.

Important decisions:

Nothing was decided on today

Action Items/Goals:

We need to finish the prototype and in the meantime group members can get started by working on the PCB.

Notes, etc:

Date: 4/6/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

Nichelle and Chris: Worked on breadboard prototype of purely transistor based design. Finished everything aside from part of the logic and the full integration.

Ishan, Arnav, Tanay: We need to finish the prototype and in the meantime group members can get started by working on the PCB.

Important decisions:

Nothing important was decided at today's meeting

Action Items/Goals:

Do the PCB and the breadboard.

Notes, etc:

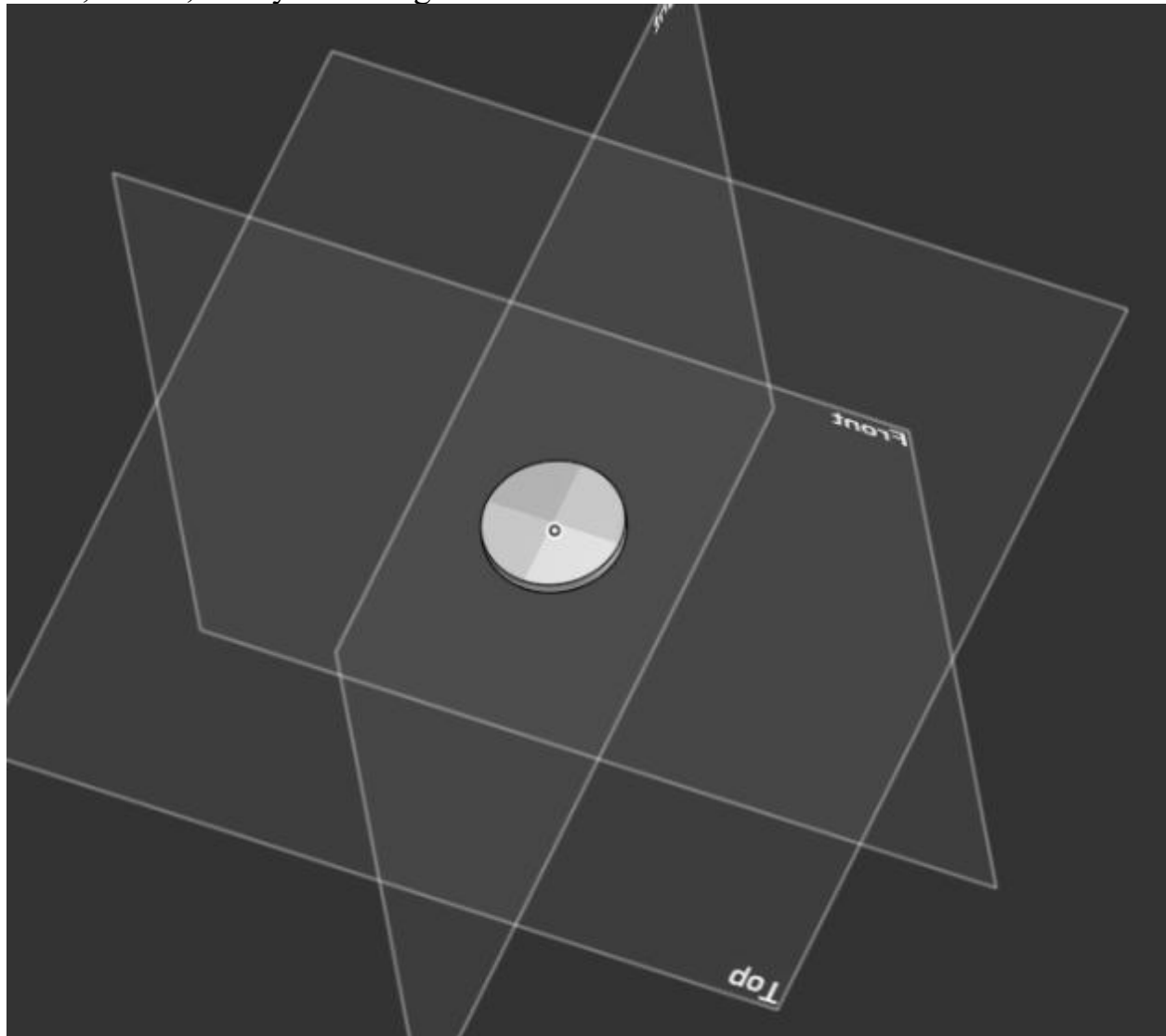
Date: 4/7/2026

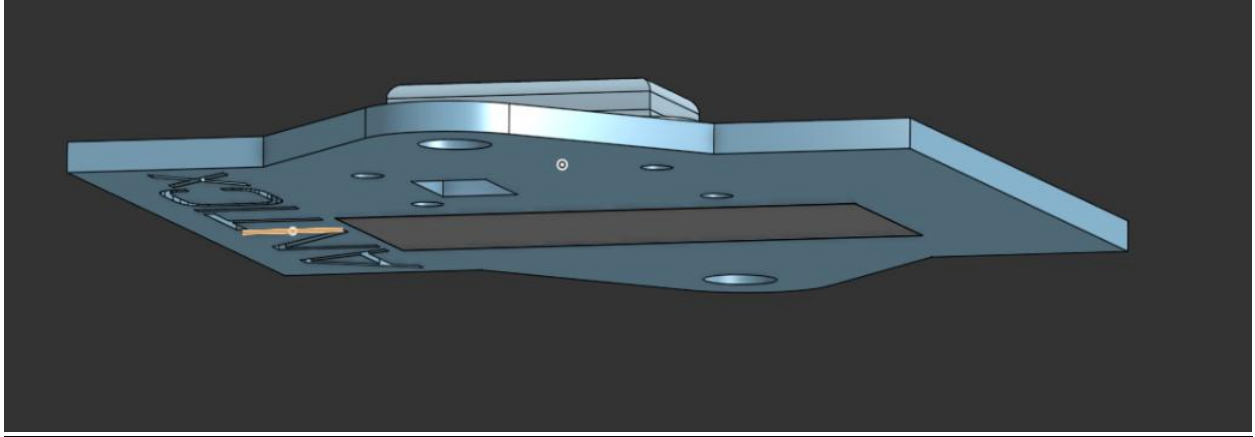
Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

Nichelle and Chris: Worked on breadboard prototype of purely transistor based design. Finished everything aside from part of the logic and the full integration.

Ishan, Arnav, Tanay: Working on the PCB.



**Important decisions:**

Nichelle, Chris, Arnav need to finish their prototype by 4/14. For the prototypes, only 1 IMU will be used.

Tanay, Ishan will start with the CAD for mounting everything on a pair of glasses

Action Items/Goals:

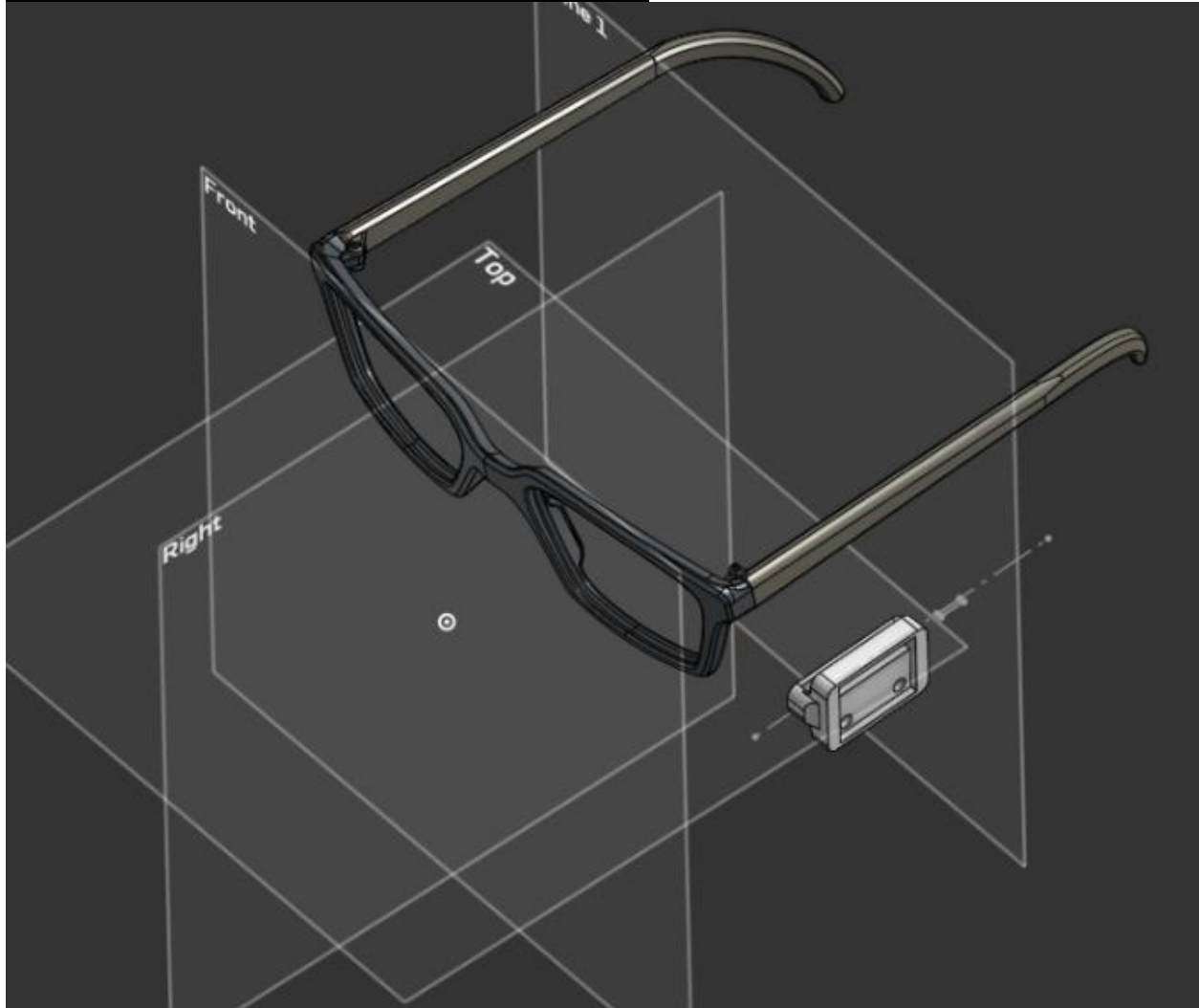
Do the PCB and the breadboard cause we kinda need to lock in.

Notes, etc:

Date: 4/8/2026

Who is Present: Ishan, Chris, Arnav, Tanav, Nichelle

What did you accomplish at the meeting?:



We CADed the glasses and IMU Stand that should be attached to said glasses.

Important decisions:

NA

Action Items/Goals:

Do the PCB and the breadboard cause we kinda need to lock in.

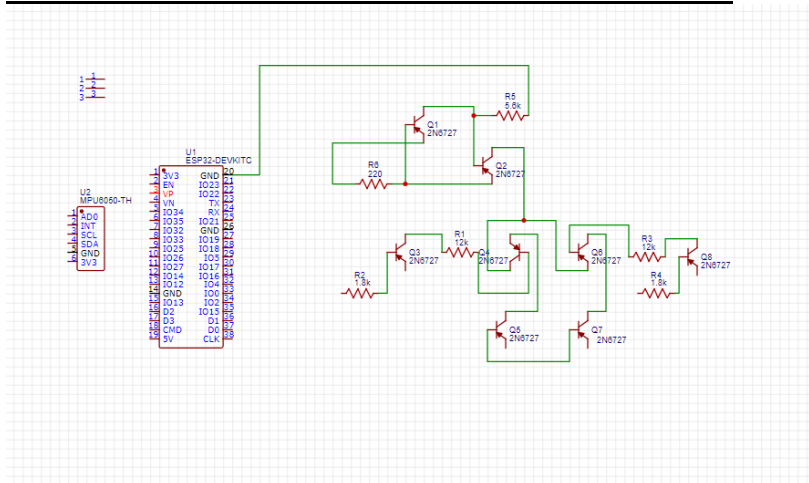
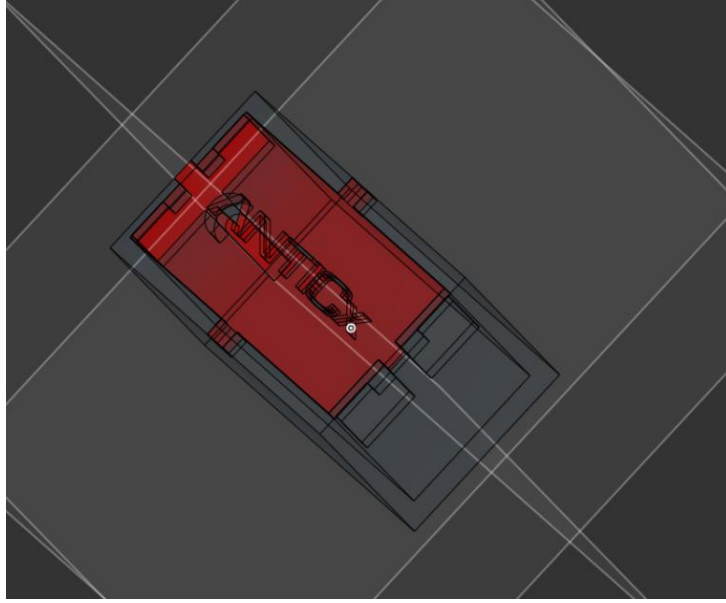
Notes, etc:

Date: 4/9/2026

Who is Present: Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

We caded the necklace holding box and we got started on the schmatic



Important decisions:

Nichelle, Chris, Arnav need to finish their prototype by 4/14. For the prototypes, only 1 IMU will be used.

Action Items/Goals:

Do the PCB and the breadboard cause we kinda need to lock in, also CAD needs to lock in and do it.

Notes, etc:

Date: 4/14/2026

Who is Present: Arnav, Tanay, Nichelle

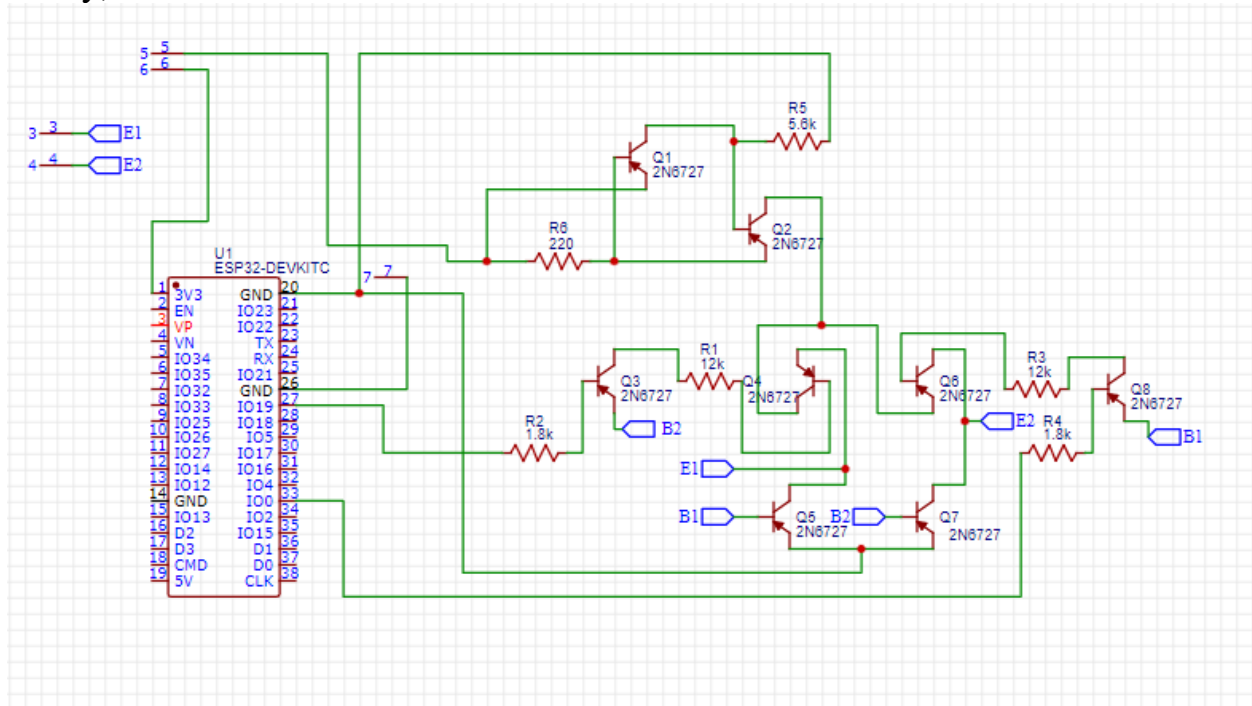
What did you accomplish at the meeting?:

Nichelle, Arnav: Worked on prototype:

- Worked on trying to connect to ESP without the boost module

Ishan: Started with CDR, finished design requirements and grading with pugh chart.

Tanay, Chris: Continued to work on the Schematic



Important decisions:

None

Action Items/Goals:

Make a schedule for creating PCBs in a timely manner, given the time constraints for design, order, and delivery.

Notes, etc:

Date: 4/15/2026

Who is Present: Arnav, Tanay, Nichelle

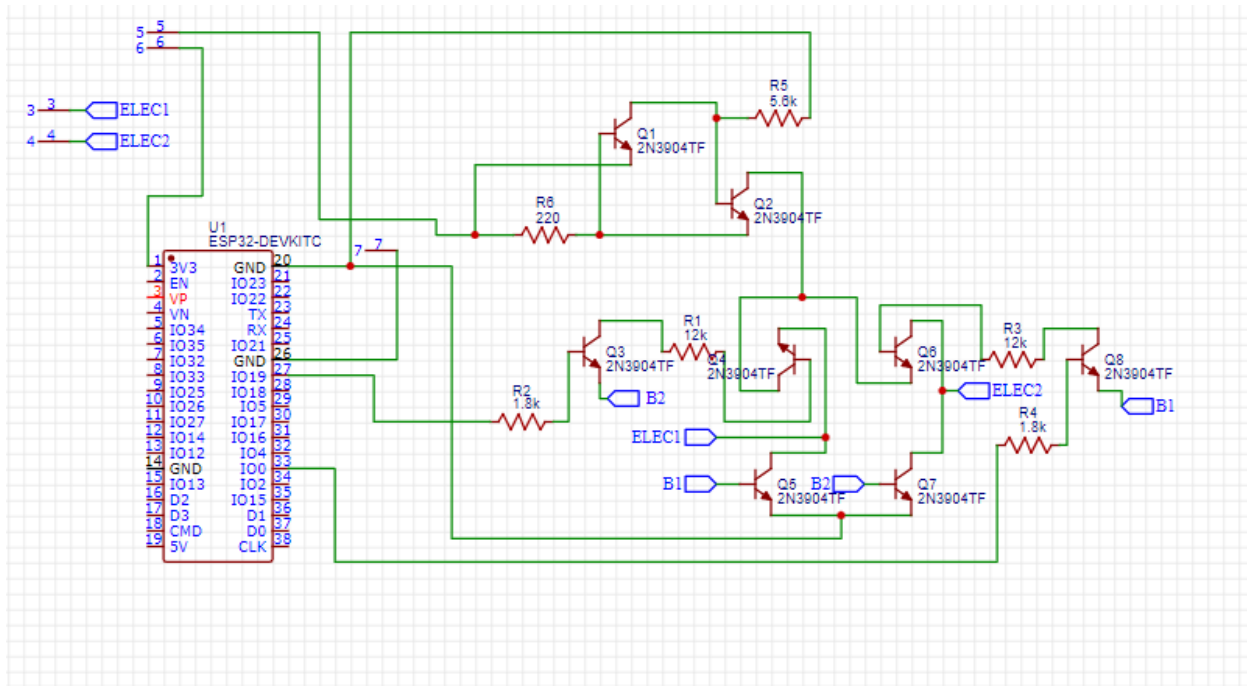
What did you accomplish at the meeting?:

Nichelle, Arnav: Worked on prototype:

- Worked on trying to connect to ESP without the boost module

Ishan: Started with CDR, finished design requirements and grading with pugh chart.

Tanay, Chris: Continued to work on the Schematic



We changed the transistors that we used cause I lowkey used the wrong one at first.

Important decisions:

None

Action Items/Goals:

Make a schedule for creating PCBs in a timely manner, given the time constraints for design, order, and delivery.

Notes, etc:

Date: 4/16/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Continued to Work on This Agenda

Nichelle, Arnav: Worked on prototype:

- Worked on trying to connect to ESP without the boost module

Ishan: Continued with CDR, finished design requirements and grading with pugh chart.

Tanay, Chris: Continued to work on the Schematic

Important decisions:

None

Action Items/Goals:

Make a schedule for creating PCBs in a timely manner, given the time constraints for design, order, and delivery.

Notes, etc:

Date: 4/17/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Continued to Work on This Agenda

Nichelle, Arnav: Worked on prototype:

- Worked on trying to connect to ESP without the boost module

Ishan: Continued with CDR, finished design requirements and grading with pugh chart.

Tanay, Chris: Continued to work on the Schematic

Important decisions:

None

Action Items/Goals:

Make a schedule for creating PCBs in a timely manner, given the time constraints for design, order, and delivery.

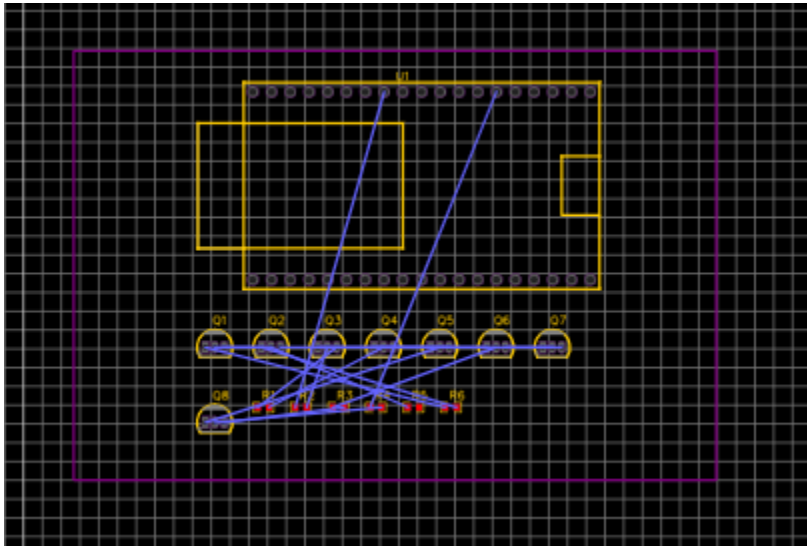
Notes, etc:

Date: 4/21/2026

Who is Present: Ishan, Chris, Arnav, Tanav, Nichelle

What did you accomplish at the meeting?:

- Tanav and Chris: Worked on PCB, finished schematic and are now making the PCB
- Ishan, Nichelle: Ordered the simpler GVS PCB and completed email template for client search
- Arnav: Worked on IMU data flow and gathered IMU data for different types of falls.



Ishan, Tanav, Arnav: Started with CDR, finished design requirements and grading with pugh chart, finalized CAD and sent it for printing

Important decisions:

Two PCBs will be tested, one with full integration and one with separation between the IMUs and the GVS system.

Action Items/Goals:

Finish larger PCB by 4/24

Notes, etc:

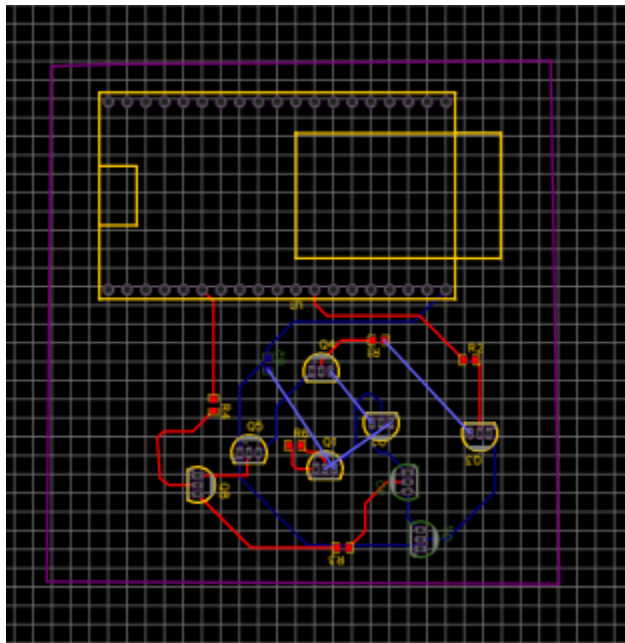
Date: 4/22/2026

Who is Present: Ishan, Chris, Arnav, Tanay, Nichelle

What did you accomplish at the meeting?:

Continued to Work on This Agenda

- Tanay and Chris: Worked on PCB, finished schematic and are now making the PCB
- Ishan, Nichelle: Ordered the simpler GVS PCB and completed email template for client search
- Arnav: Worked on IMU data flow and gathered IMU data for different types of falls.



Ishan, Tanay, Arnav: Started with CDR, finished design requirements and grading with pugh chart, finalized CAD and sent it for printing

Important decisions:

Two PCBs will be tested, one with full integration and one with separation between the IMUs and the GVS system.

Action Items/Goals:

Finish larger PCB by 4/24

Notes, etc:

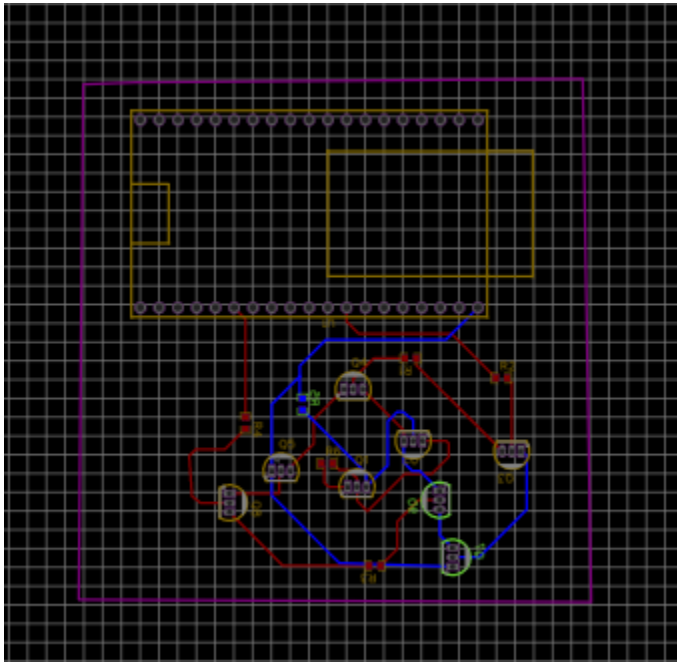
Date: 4/23/2026

Who is Present: Ishan, Chris, Arnav, Tanav, Nichelle

What did you accomplish at the meeting?:

Continued to Work on This Agenda

- Tanav and Chris: Worked on PCB, finished schematic and are now making the PCB
- Ishan, Nichelle: Ordered the simpler GVS PCB and completed email template for client search
- Arnav: Worked on IMU data flow and gathered IMU data for different types of falls.



Ishan, Tanav, Arnav: Started with CDR, finished design requirements and grading with pugh chart, finalized CAD and sent it for printing

Important decisions:

Two PCBs will be tested, one with full integration and one with separation between the IMUs and the GVS system.

Action Items/Goals:

Finish larger PCB by 4/24

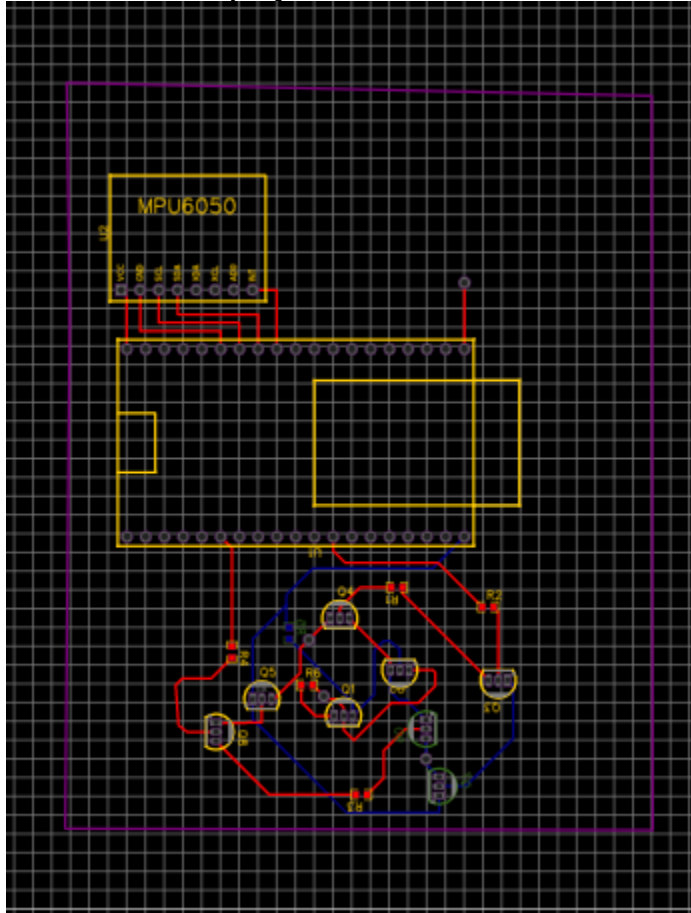
Notes, etc:

Date: 4/24/2026

Who is Present: Ishan, Chris, Arnav, Tanav, Nichelle

What did you accomplish at the meeting?:

We worked on integrated the MPU 6050 into the PCB and seeing how something like that could play out.



Important decisions:

NA

Action Items/Goals:

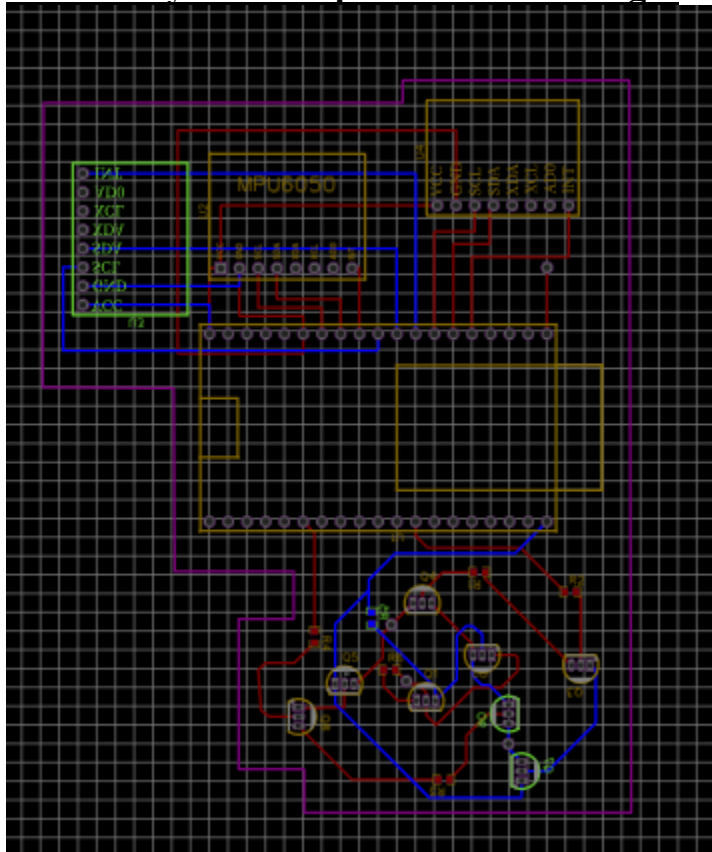
See how to integrate the 2 more IMUs and the pins for that.

Notes, etc:

Date: 4/27/2026

Who is Present: Ishan, Chris, Arnav, Tanav, Nichelle

What did you accomplish at the meeting?:



Figured out how to add two more IMUs to the PCB and how to manage all the pins, everything as to be checked over one final time.

Important decisions:

NA

Action Items/Goals:

NA

Notes, etc:

Date: 4/29/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Replaced boost converter and re-verified charge threshold

Conducted multimeter-based testing of electrodes

Updated P.O.C. doc with images

Re-soldered IMU pins

Important decisions:

If breadboard circuit has to be used as a fallback, will likely recreate circuit on a larger board

Action Items/Goals:

Ensure prototype functionality and safety

Notes, etc:

N/A

Date: 4/30/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Updated POC background
Created and tested arduino code with multimeter
Helped other groups with electronics

Important decisions:

Arduino should not be used as the final microcontroller

Action Items/Goals:

Ensure prototype functionality and safety

Notes, etc:

N/A

Date: 5/01/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Updated POC sources and proofread
Helped other groups with electronics

Important decisions:

Electrodes need a better adhesive or glasses design should ensure consistent surface contact in other ways

Action Items/Goals:

Ensure prototype functionality and safety

Notes, etc:

N/A

Date: 5/04/2026

Who is Present: All of Us

What did you accomplish at the meeting?:

During this time period, Arnav continued to code the device with the fall detection accuracy, doing testing while coding to make sure the results were making sense.

Nichelle and Chris were doing testing with the GVS system separately.

Ishan and Tanay were working on the Design study document

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/05/2026

Who is Present: All of Us

What did you accomplish at the meeting?:

During this time period, Arnav continued to code the device with the fall detection accuracy, doing testing while coding to make sure the results were making sense.

Nichelle and Chris were doing testing with the GVS system separately.

Ishan and Tanay were working on the Design study document

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/06/2026

Who is Present: All of us

What did you accomplish at the meeting?:

During this time period, Arnav continued to code the device with the fall detection accuracy, doing testing while coding to make sure the results were making sense.

Nichelle and Chris were doing testing with the GVS system separately.

Ishan and Tanay were working on the Design study document

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/08/2026

Who is Present: All of us

What did you accomplish at the meeting?:

During this time period, Arnav continued to code the device with the fall detection accuracy, doing testing while coding to make sure the results were making sense.

Nichelle and Chris were doing testing with the GVS system separately.

Ishan and Tanay were working on the Design study document

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/11/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Ishan and Tanay worked on the assignment for the AT-PROJECT- Project Title, Description, and LOGO

Nichelle, Arnav, and Chris worked on testing the device together

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/12/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Ishan and Tanay worked on the poster

Nichelle, Arnav, and Chris worked on testing the device together

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/13/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Ishan and Tanay worked on the poster

Nichelle, Arnav, and Chris worked on testing the device together

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/14/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Tanay worked on the handout

Ishan worked on the Design Study Document

Nichelle, Arnav, and Chris worked on mounting the device and testing comfort and some of the requirements

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/18/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Tanay worked and finished the handout with input from Nichelle, Arnav, and Chris
Ishan worked on the Design Study Document

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/19/2026

Who is Present: Nichelle

What did you accomplish at the meeting?:

Tanay and Ishan helped other groups with projects

Nichelle, Chris, and Arnav worked on the device and making final checks before the fair.

Important decisions:

Action Items/Goals:

Finish the tasks that were assigned

Notes, etc:

N/A

Date: 5/20/2026

Who is Present: Everyone

What did you accomplish at the meeting?:

The fair!

Important decisions:

N/A

Action Items/Goals:

Have fun and explain our project to the community

Notes, etc:

N/A

All Ideas Here

- Device that amplifies noise (hearing aids) based on the direction that the old person is facing
- Something that is like a doormat which can help people to not forget stuff when they leave their house.
- A phone case which corrects for tremors, more to it but all I can type.
- **Potential Idea: The system uses camera and IMU data to predict balance instability in older adults and applies low-level galvanic vestibular stimulation to bias postural perception and reduce fall risk, rather than attempting real-time physical correction**
- Buttons in real life that correspond to buttons on the phone
- Old people physical therapy trainer
- Phone case that makes text appear more visible for old people
- Shape adapting attachment for handles for old people with arthritis
- Slippers/shoes that slightly adjust with tiny actuators to make sure people do not lose balance
- Cup/mug that doesn't spill when tremors
- Inexpensive but more mobile soft-exterior therapy robot
- ADHD-responsive study environment
- Anxiety-responsive wristband or glove for emotional comfort
- Center-of-gravity-adjusting backpack for balance assistance

Tanay (Final Idea): Galvanic Vestibular Stimulation sleep posture system — uses GVS during sleep to retrain vestibular sensitivity in older adults, since the vestibular system remains partially active during sleep and passive stimulation could have rehabilitative effects over time.